Ledo Thankachan | Healthcare Data Analyst

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SUMMARY

Data professional with over 3+ years of experience in programming, data engineering, and advanced analytics. Skilled in Python, R, and SQL for large-scale data analysis and modeling. Experienced in building ETL pipelines, optimizing data warehouses, and implementing dimensional models using platforms such as Azure Synapse and Databricks. Proficient in creating interactive dashboards with Tableau and Power BI to deliver actionable insights. Strong background in predictive modeling, Bayesian statistics, and risk stratification with a focus on accuracy, compliance, and data governance.

TECHNICAL SKILLS

Programming & Data Analysis: Python (pandas, NumPy, scikit-learn), R, SQL (PostgreSQL, MySQL, SQL Server), Excel (PivotTables, nested formulas, dashboards)
Data Engineering & Modeling: ETL (Extract, Transform, Load), Data Warehousing, Dimensional Data Modeling, Azure Synapse, Databricks
Visualization & BI Tools: Tableau, Power BI, RxConnect (pharmacy data system), Excel Analytics
Statistical & Analytical Methods: Regression, Bayesian Statistics, Hypothesis Testing, Predictive Modeling, Risk Stratification, Clinical Outcomes Analysis
Healthcare Data Standards: HL7, FHIR, HIPAA, HEDIS, CMS Star Ratings, NHANES, All of Us Research Program datasets
Data Quality & Governance: Data Validation, Anomaly Detection, Data Lineage, Version Control, Reproducibility Practices

EXPERIENCE

Pharmacy Technician **Aug 2025 - Present**
CVS Health | Fairfax, VA

- Processed 150+ prescriptions daily with 99% accuracy, ensuring compliance with HIPAA and state/federal regulations.
- Conducted medication reconciliation and insurance claim adjudication, resolving prior authorization and billing issues to reduce patient delays by 20%.
- Monitored and managed inventory levels of 1,000+ medications, reducing stockouts and wastage through timely ordering and tracking.
- Utilized RxConnect and internal CVS data systems to streamline prescription entry, workflow management, and insurance claims.

Graduate Research Assistant/Graduate Teaching Assistant **Aug 2023 – May 2025**
George Mason University – Fairfax, VA

- Managed and analyzed large-scale clinical datasets using Python, SQL, and R for NIH-aligned studies, ensuring data accuracy, reproducibility, and compliance with HIPAA and IRB guidelines.
- Abstracted and processed clinical data from EHR systems and national datasets (e.g., NHANES), producing reports and visualizations for principal investigators.
- Conducted literature reviews and protocol documentation for IRB submissions and manuscript development, adhering to GCP standards.
- Collaborated with cross-disciplinary teams to ensure data integrity, version control, and alignment with study goals and federal regulations.
- Mentored students in healthcare data analysis using Python, Pandas, and Bayesian statistics.
- Supported faculty-led research on bipolar disorder outcomes using the All of Us Research Program dataset, applying Bayesian inference to assess risk factors and treatment response across diverse populations.
- Guided over 60 students in mastering Excel analytics including dashboards, PivotTables, and nested formulas.

Healthcare Data Analyst **Sep 2021 – Nov 2022**
Citius Tech | India

- Collected and integrated data from EHR systems, claims platforms, and medical devices, ensuring compliance with HIPAA and FHIR standards while improving data availability across teams.
- Built and optimized data models in Azure Synapse and Databricks, reducing query run times by 30% and enabling faster turnaround for clinical and operational reporting.
- Developed predictive models to flag patients at high risk of hospital readmission, supporting care management programs that helped lower projected readmissions by 15%.

- Designed interactive dashboards in Power BI and Tableau that tracked denial rates, clinical outcomes, and cost drivers, cutting manual reporting time from days to hours.
- Partnered with clinicians, operations staff, and IT teams to define analytics requirements, ensuring insights were clear, actionable, and aligned with day-to-day healthcare workflows.
- Implemented automated data quality checks that reduced reporting errors by 25%, strengthening the reliability of compliance and performance dashboards.

EDUCATION

Masters in Health Informatics **Dec 2024**
George Mason University, Fairfax, VA, USA
Bachelors in Doctor of Pharmacy (PharmD) **Sep 2022**
JNTUH, Hyderabad, India

PROJECTS

Capstone: Predictive Modeling of ER Admissions

- Built a Python-based model using EMR and claims data to predict emergency room visits with 75% accuracy, aiding care coordination strategies.
- Automated SQL pipelines for real-time data ingestion, validation, and transformation across multiple care settings.

MyCare IQ Web Tool

- Designed a zip-code-based provider and insurance ranking tool using Python, HTML, and public insurance datasets, supporting patient decision-making.

Predictive Modelling for Stroke Risk Assessment

- Conducted comprehensive data cleaning, standardizing processes to ensure data integrity and consistency.
- Developed machine learning models (logistic regression, random forest, decision tree, neural network, ADA boosting machines) to predict stroke risk using python libraries such as scipy, sklearn, Keras, TensorFlow.
- Evaluated model performance using metrics like accuracy, precision, recall, and AUC-ROC.

Readmission rate analysis in Benefis Hospital

- Developed interactive Tableau dashboards to analyze readmission rates at Benefits Hospital, utilizing data visualizations such as scatter plots, bar graphs, and heat maps.
- Identified correlations between readmission rates and patient demographics, diagnoses, treatments, and geographical factors.

Trends in smoking before and after the year 2000

- Conducted comprehensive descriptive, univariate, and multivariate analysis of smoking data using Python, focusing on trends before and after 2000.
- Created an interactive website using HTML to present the analyzed smoking trend data, facilitating user exploration of the findings and insights.
- Developed Python scripts using matplotlib, pandas, and numpy library to do statistical analysis and generate various visualizations found smoking cigarettes was more before the year 2000.

Investigating the Role of Tocilizumab, Dexamethasone, and Methylprednisolone in Severe COVID-19 Cases

- Conducted real-time data analysis using SPSS to assess the efficacy of Tocilizumab, Dexamethasone, and Methylprednisolone in severe COVID-19 cases, performing statistical tests including mean, median, and ANOVA to compare treatment outcomes.
- Visualized findings through graphs for effective communication of results using SPSS.

PUBLICATIONS

- Ethambutol optic neuropathy in the extended anti-tubercular therapy regime: A systematic review. (Verify)
- Association of lower anti spike antibody levels with mortality in ICU patients with COVID-19 Disease. (Verify)
- Duration of immunity against COVID-19 after vaccination in the Indian subcontinent. (Verify)
- A case presentation on Ofloxacin induced dermal hypersensitivity reaction. (Verify)